

Rate Limit and Donation Plan

For azielcorpuslibrary.net only

RL-WP-0.1 (library scope) 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local specification and cost plan for the Digital Library Worker. Origin: <https://www.azielcorpuslibrary.net> Evidence window: Cloudflare billable usage, 2–9 September 2026 (8 of 30 cycle days).

0. Sentence

This plan is for the library. Not godlock.uk. Not the software tabs. Not local ChainLock. Visitors can still search and open cards. Bots cannot walk every AZDOC key. The operator is uncapped.

1. Scope

In scope:

- <https://www.azielcorpuslibrary.net>
- `/v1/search`
- record cards (AZDOC id, title, content_sha256, chain_tip, shelf)
- the Worker KV that backs those cards
- a Donate strip on the library origin

Out of scope:

- godlock.uk
- azieleliab.com software tabs
- aziel-runtime Worker CPU as a separate product
- local chainlock.py vaults
- PDF bytes (those stay on the object shelf; sync already does not pull PDFs)

ChainLock library-sync is a client of `/v1/search`. After this plan it should hit one packed index, not 191 keys.

2. What the bill said

Observed 2–9 September 2026 on the Cloudflare account that serves the library:

- Total cost: \$142.26
- Projected cycle: \$533.47 (2 September – 1 October)
- Average daily: \$17.78
- KV Read Operations: 271.91M — **\$131.00**
- KV List Operations: 1.33M — \$5.00
- KV Write Operations: 1.37M — \$5.00
- Workers CPU ms: 92.36M — \$1.26
- Workers Standard Requests: 1.42M — inside the 10M included

- D1 rows read: 5.44B — inside the 25B included (\$0)

First 10 million KV reads per month are included, then \$0.50 per million. \$131 is 262 million billable reads. That matches 271.91M minus the included 10M.

1.42 million requests / 271.91 million KV reads $\approx$ **191 KV operations per library request**. People are not opening 272 million pages. The library hot path is listing or getting too many keys per search or per record view.

3. Claim

One packed library index, edge cache on `/v1/search` and record cards, a per-visitor cap on the library Worker, and a server-side operator token cut billed KV reads without taking the shelf down.

4. What it is not

- Not a public Node Gate panel on the library.
- Not an IP allow/block UI.
- Not a deletion of AZDOC records.
- Not a change to ChainLock's offline `library.jsonl` shelf.
- Not invented wallet addresses.
- Not rate limits on GodLock or on the operator's local runtime.

5. Root cause on this origin

Likely hot-path habits on the library Worker:

- `KV.list()` of the AZDOC namespace on every search
- one `KV.get` per hit instead of one packed search result
- missing `cacheTtl` on card gets
- scrapers walking `/record/AZDOC-*` and `/v1/search?q=` across the catalog
- wrangler / dashboard list also bills

D1 already serves 5.44 billion row reads inside the free included band. If search can run on D1 or on one cached JSON index, KV stops being the catalog.

6. Plan, library only

Step 1 — Pack the library index. One key `library:index:v1`: AZDOC id, title, shelf, `content_sha256`, `chain_tip`, updated ts. No PDF body. `/v1/search` filters that object in memory or in D1. `KV.list()` leaves the request path.

Step 2 — Cache cards. `KV.get(id, { cacheTtl: 3600 })` for a single record. Search responses: `Cache-Control: public, s-maxage=120, stale-while-revalidate=3600`. Repeat identical queries should not bill KV.

Step 3 — Cap KV work per library request. At most 8 KV operations per invocation (index + one card + receipts). Over that: return packed hits and truncated. Do not walk the namespace.

Step 4 — Per-visitor bucket on this hostname only. Visitor id = SHA-256 of CF-Connecting-IP plus optional cookie. Host match: `azielcorpuslibrary.net` only.

Default:

- 30 search requests / minute
- 120 record views / hour
- 800 library requests / day

Soft exceed: 429, Retry-After: 30, body is the last cached search page so the shelf still appears up. Hard isolate: SweepGate poison / scraper only. Quarantine. No merge into ChainLock STM.

Step 5 — Operator exclude. Secret header or Cloudflare Access service token in server-side `gate_config.json`. Operator sync, ingest, and verify skip the bucket. Same token must not appear in the library UI.

Step 6 — Cycle alert. Budget cap \$80–\$100 on this account so a catalog scrape cannot run to \$533.

Step 7 — Measure on this origin. Success: KV reads per library request from ~191 toward 1–2. `/v1/search` stays up. `library-sync --query Q` still returns AZDOC cards. Offline `library.jsonl` unchanged.

7. What a limited visitor still gets

- search within quota
- last cached result page when soft-limited
- record cards they already opened
- PDF links as they exist today (no new PDF pull into KV)
- ChainLock local shelf if they synced earlier

They lose unbounded crawl of every AZDOC key.

8. Donation tab on the library

Static island on `azielcorpuslibrary.net` only for this paper.

- No Worker. No KV.
- Networks the operator already controls. Paste addresses at publish. This paper does not invent them.
- Copy plus optional public-plane tx note.
- No legal name. No home. No case narrative.

The tab funds the cycle cap. It does not replace Steps 1–4.

9. Wire on the library Worker

`frag → sweep → [library-bucket] → fold → static → packed index / card → emit`

Refuse: `sweep-isolate` | `rate-soft` | `rate-hard` | `kv-cap`. ChainLock clients treat `rate-soft` as “use local shelf.” Site-fail was already the point of `library.jsonl`.

10. Cap

Library plan only. Do not copy these buckets onto GodLock or onto local ChainLock. Operator token stays off the page.

Companion papers: CL-WP-0.4 (library tether), AP-WP-0.2, SG-WP-0.1, LS-WP-0.1.